Statistical Methods (MA2220)

Project Report

Statistical Decoding and Forecasting of India's Multidimensional Poverty Index (MPI) Using NFHS-5 Data

Submitted by:

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Abstract

Poverty is a complex and multidimensional issue that extends far beyond the lack of income. Factors such as poor health, limited education, inadequate housing, lack of sanitation, malnutrition, and restricted access to basic services collectively shape the quality of life of individuals and communities. Traditional poverty measures based solely on income often fail to capture these interconnected deprivations. To address this limitation, the Multidimensional Poverty Index (MPI) provides a broader framework for understanding poverty through indicators related to health, education, and living standards.

This study, titled "Statistical Decoding and Forecasting of India's Multidimensional Poverty Index (MPI) Using NFHS-5 Data," analyzes multidimensional poverty across Indian districts using data obtained from the National Family Health Survey (NFHS-5) conducted during 2019–2021. The study combines statistical methods and machine learning approaches to identify important socio-economic factors influencing poverty and to forecast deprivation patterns across districts. The project utilizes indicators such as literacy, school attendance, sanitation, electricity access, clean cooking fuel, child malnutrition, and healthcare access to construct and analyze a proxy MPI score. Exploratory data analysis, statistical modeling, and machine learning techniques including Decision Trees, Random Forests, and K-Means Clustering are employed to understand relationships among poverty indicators and evaluate predictive performance.

The findings highlight that education and sanitation are among the strongest determinants of multidimensional poverty, reinforcing the idea that poverty reduction requires integrated social and policy interventions rather than isolated economic measures. The study demonstrates the practical relevance of MPI in identifying vulnerable regions, supporting evidence-based policymaking, and enabling more targeted welfare strategies in India.

1 Introduction

1.1 Background of the Study

India has experienced rapid economic growth over the past few decades; however, significant inequalities continue to exist across regions, communities, and social groups. While income levels have improved for many households, millions of people still face deprivation in areas such as healthcare, nutrition, education, sanitation, housing, and access to essential services. These forms of deprivation directly affect human development and quality of life.

Traditionally, poverty has been measured using income or consumption expenditure. Although such measures are useful, they fail to fully represent the actual living conditions of people. A family may earn above the poverty line but still suffer from malnutrition, lack of education, poor sanitation, or inadequate healthcare. Therefore, poverty cannot be understood solely through financial indicators.

This limitation led to the development of the Multidimensional Poverty Index (MPI), which evaluates poverty using multiple dimensions of human well-being. MPI provides a broader and more realistic understanding of deprivation by considering indicators related to health, education, and living standards.

By applying statistical analysis and machine learning techniques, this project aims to identify the major drivers of poverty and forecast deprivation patterns across regions. The study also demonstrates how data-driven approaches can support policymakers in designing more targeted and effective poverty alleviation strategies.

1.2 Understanding Poverty Beyond Income

Poverty is commonly associated with low income or lack of money. However, in reality, poverty is a much broader concept that affects multiple aspects of human life. A person may be considered poor not only because they earn less, but also because they:

- Lack access to quality education
- Suffer from malnutrition or poor healthcare
- Live in unsafe or inadequate housing conditions
- Do not have access to sanitation, electricity, or clean water
- Face social exclusion and limited opportunities

Thus, poverty represents a condition where individuals are deprived of the basic capabilities and opportunities required to live a healthy, dignified, and productive life.

1.3 What is the Multidimensional Poverty Index (MPI)?

The Multidimensional Poverty Index (MPI) is a measure developed to assess poverty using multiple indicators of human well-being instead of relying solely on income. MPI evaluates deprivation across three major dimensions:

1. Health This dimension examines whether individuals have access to proper nutrition and healthcare. Indicators often include:
   - Child malnutrition
   - Child malnutrition
   - Stunting
   - Anemia
   - Access to healthcare services
2. Education This dimension measures educational access and attainment through indicators such as:
   - Literacy
   - School attendance
   - Years of schooling
3. Living Standards This dimension evaluates the quality of household living conditions using indicators like:
   - Access to electricity.
   - Sanitation facilities
   - Availability of drinking water
   - Clean cooking fuel
   - Housing conditions

In this study, indicators from the NFHS-5 data were classified into these three dimensions to construct and analyze a proxy MPI score. Unlike traditional poverty measures, MPI identifies how people are poor and which dimensions contribute the most to deprivation. This makes MPI especially useful for designing targeted welfare policies and developmental programs.

1.4 Importance of MPI in India

MPI is particularly important in India because poverty in the country is highly diverse and region-specific. Different districts and states face different forms of deprivation. India's large population and socio-economic diversity make multidimensional analysis essential for effective policymaking.

MPI captures multiple aspects of deprivation instead of focusing only on income levels. It helps evaluate the effectiveness of welfare schemes such as Poshan Abhiyaan, Swachh Bharat Mission, Pradhan Mantri Ujjwala Yojana, National Health Mission et cetera. It also enables for comparison between districts and states, helping identify high-deprivation regions that require urgent intervention.

Data Description (NFHS-5)

Data Source and Overview

The dataset used in this study is obtained from the National Family Health Survey (NFHS-5), conducted during 2019–2021 across India. The survey provides comprehensive district-level information on various indicators related to health, education, and living standards.

The data is cross-sectional in nature and consists of aggregated indicators at the district level, enabling comparative analysis across different regions of the country.

Structure of the Data

The dataset contains observations corresponding to districts, with each row representing a district and each column representing a specific socio-economic or health indicator. The variables are primarily expressed as percentages, making them suitable for comparative and statistical analysis.

Classification of Variables

For the purpose of this study, a subset of relevant variables is selected and classified into three major dimensions inspired by the Multidimensional Poverty Index (MPI): health, education, and living standards.

Dimension	Variable	Description
Health	Stunting	Children under 5 years who are stunted (chronic malnutrition)
Health	Underweight	Children under 5 years who are underweight
Health	Anaemia	Children age 6–59 months who are anaemic
Health	Institutional births	Births occurring in health facilities indicating healthcare access
Education	School attendance	Female population age 6+ who ever attended school
Education	Literacy	Women (age 15–49) who are literate
Education	Years of schooling	Women with 10 or more years of schooling
Education	Pre-primary education	Children attending pre-primary school
Living Standard	Electricity	Households with access to electricity
Living Standard	Drinking water	Households with improved drinking water source
Living Standard	Sanitation	Households with improved sanitation facilities
Living Standard	Clean fuel	Households using clean cooking fuel
Living Standard	Health insurance	Households covered under health insurance schemes
Living Standard	Iodized salt	Households using iodized salt

Description of Key Variables

The selected variables capture multiple aspects of deprivation. Indicators such as stunting, underweight, and anaemia reflect the nutritional and health status of children. Institutional births serve as an indicator of access to healthcare services.

Education-related variables such as literacy, school attendance, and years of schooling capture both access to and attainment of education, which are crucial determinants of socio-economic development.

Living standard indicators such as electricity, sanitation, drinking water, and clean cooking fuel reflect the quality of household infrastructure and overall well-being.

Preliminary Observations

A preliminary examination of the dataset reveals significant variation across districts in terms of health, education, and living standard indicators. In particular, districts with lower access to basic amenities such as sanitation and clean cooking fuel often exhibit poorer health outcomes, including higher levels of child malnutrition.

This structured understanding of the dataset forms the foundation for subsequent preprocessing, statistical modeling, and machine learning analysis carried out in this study.

2 Data Preprocessing

The dataset was preprocessed to ensure accuracy, consistency, and suitability for analysis. The following steps were carried out:

2.1 Data Import and Removal of Non-Numeric Columns

The dataset was imported into RStudio from an Excel file. Columns such as district and state names were removed, as they act only as identifiers and are not required for numerical analysis.

2.2 Data Cleaning and Handling Missing Values

The dataset contained values with brackets and special characters (e.g., (29.5)), which were cleaned by removing non-numeric characters. Missing values were then handled using mean imputation, where each missing entry was replaced with the mean of its respective column. This approach preserves the dataset size and prevents data loss.

2.3 Creation and Export of Clean Dataset

After preprocessing, the cleaned dataset was stored in a new dataframe (`df_clean`). It was then exported as a CSV file (`stats_clean_data.csv`) for further analysis and visualization.

3 Exploratory Data Analysis

3.1 Histogram Analysis

3.1.1 Literacy Distribution

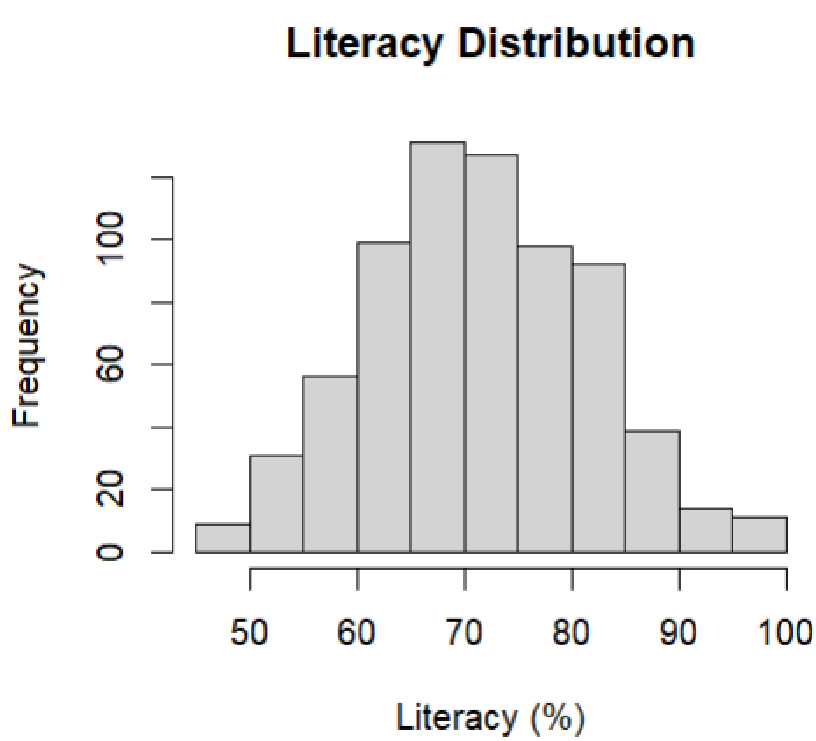

Figure 1: Histogram of Literacy Distribution

Interpretation:
The histogram of literacy shows a roughly bell-shaped (normal-like) distribution, with most districts concentrated between 60% and 80%. The peak around 65–75% indicates that a majority of districts have moderate to high literacy levels. The relatively small number of districts at the lower and higher extremes suggests that very low or very high literacy rates are less common, indicating a fairly balanced distribution with moderate variability.

3.1.2 Child Malnutrition Distribution

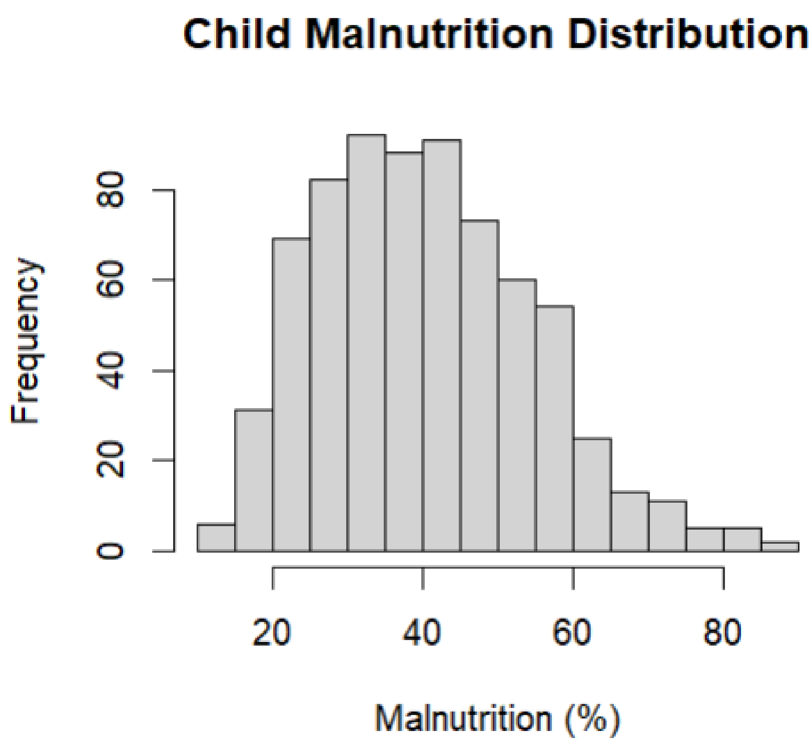

Figure 2: Histogram of Child Malnutrition

Interpretation:
The distribution of child malnutrition appears right-skewed, with most districts concentrated in the lower to mid range (around 20%–50%), and fewer districts showing very high malnutrition levels. The long tail towards higher values indicates that while most regions have moderate malnutrition, some districts experience significantly worse conditions. This suggests inequality in health outcomes across regions.

Insight:
"While literacy levels are relatively evenly distributed across districts, malnutrition shows greater skewness and variability, indicating uneven health outcomes despite similar educational levels."

3.2 Scatter Plot Analysis

3.2.1 Literacy vs Malnutrition

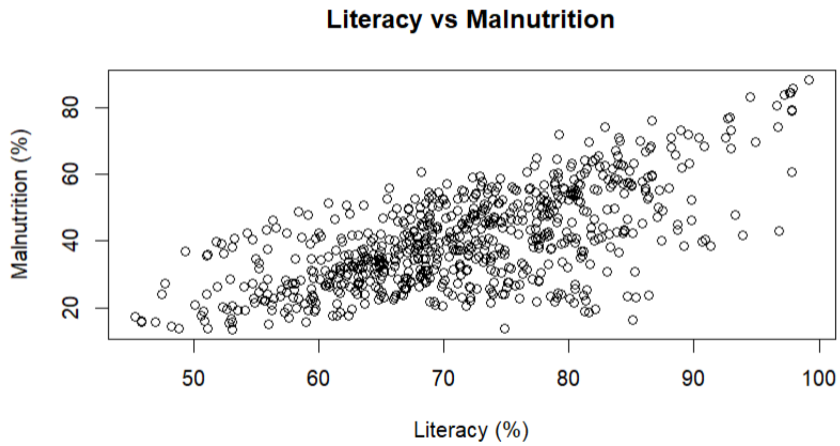

Figure 3: Scatter Plot of Literacy vs Malnutrition

Interpretation:
The scatter plot shows a positive relationship between literacy and malnutrition, as the points trend upward. This is somewhat unexpected, as higher literacy is generally associated with better health outcomes. However, the wide spread of data points suggests that literacy alone does not fully explain variations in malnutrition. Other factors such as income, healthcare access, and regional disparities may play a significant role. Hence, the relationship appears moderate but not strongly linear, indicating the multidimensional nature of poverty.

3.2.2 Sanitation vs Underweight

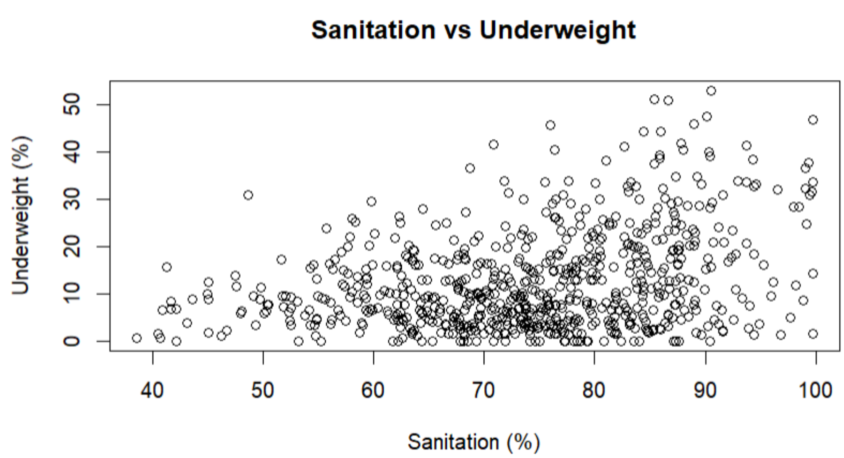

Figure 4: Scatter Plot of Sanitation vs Underweight

Interpretation:
The scatter plot between sanitation and underweight children does not show a clear linear relationship. While sanitation levels increase, underweight percentages vary widely, indicating high dispersion in the data. Although some districts with better sanitation have lower underweight levels, the presence of many scattered points suggests that sanitation alone is insufficient to explain nutritional outcomes. This highlights that multiple factors contribute to undernutrition, including food availability and healthcare access.

Insight:
"The scatter plots indicate that individual indicators such as literacy and sanitation do not independently determine health outcomes, reinforcing the multidimensional nature of poverty."

3.3 Boxplot Analysis

3.3.1 Literacy Spread

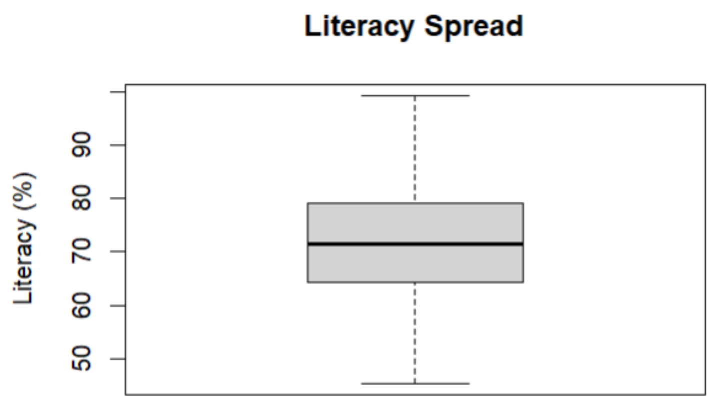

Figure 5: Boxplot of Literacy

Interpretation:
The boxplot of literacy shows that most districts have literacy rates concentrated around the median value of approximately 70%. The interquartile range indicates moderate variability, suggesting that while literacy is relatively consistent across many districts, there are still noticeable differences. The absence of extreme outliers implies that literacy levels are fairly evenly distributed, with most districts falling within a similar range.

3.3.2 Underweight Distribution

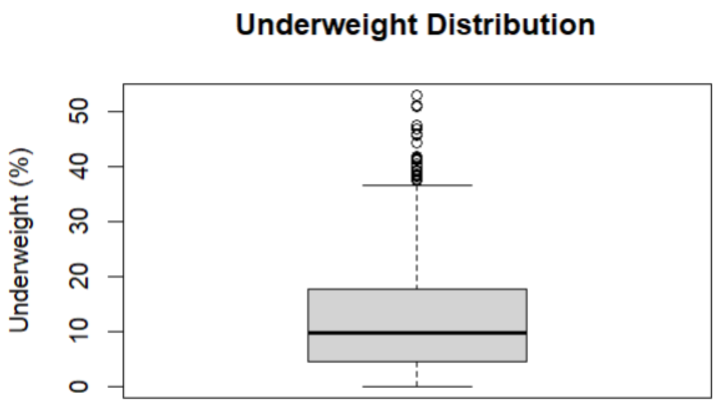

Figure 6: Boxplot of Underweight Children

Interpretation:
The boxplot for underweight children shows a median around a lower percentage, but with a wide spread of values. The presence of several high-value outliers indicates that certain districts experience significantly higher levels of undernutrition compared to others. This suggests considerable inequality in nutritional outcomes, with some regions facing severe health challenges despite overall moderate levels.

Insight:
"The boxplots highlight variability and inequality across districts, particularly in health indicators, reinforcing the uneven distribution of poverty-related outcomes."

3.4 Correlation Analysis (Heatmap)

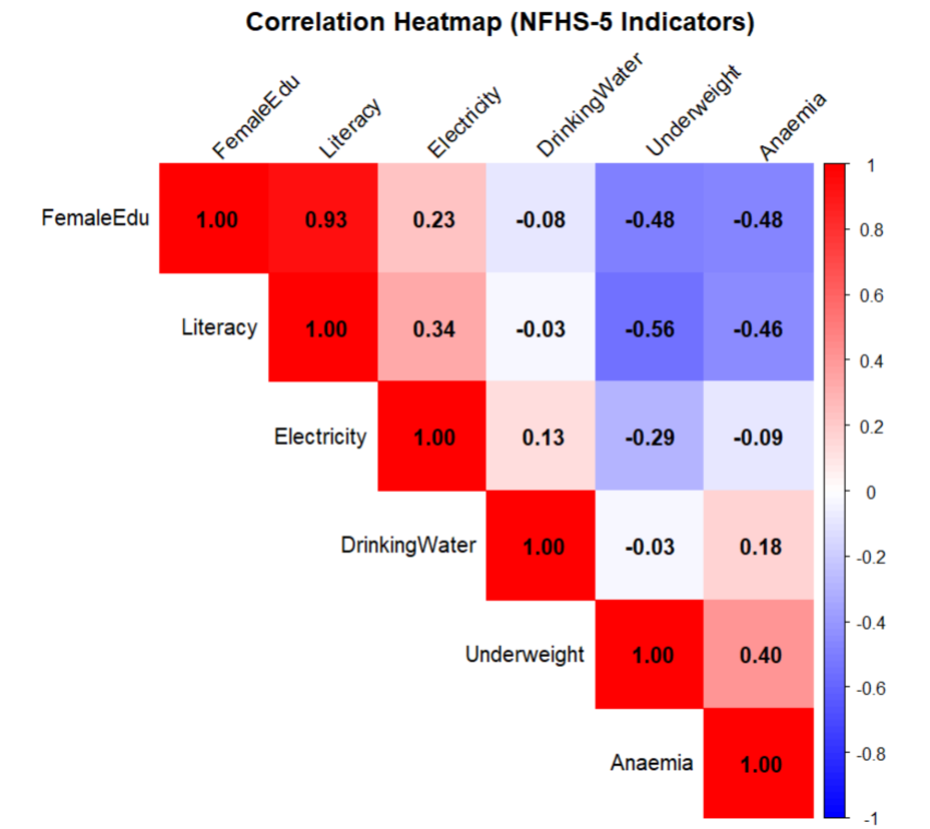

Figure 7: Correlation Heatmap of Variables

Interpretation:
The correlation heatmap shows the relationships between selected NFHS-5 indicators. Red shades represent positive correlations, while blue shades represent negative correlations. Darker colors indicate stronger relationships.

A very strong positive correlation is observed between Female Education and Literacy (0.93), indicating that districts with higher female education levels generally have higher literacy rates. Health indicators such as Underweight and Anaemia (0.40) also show a moderate positive relationship, suggesting that regions with higher undernutrition tend to have higher anaemia levels.

Negative correlations are observed between education indicators and health-related variables. For example, Literacy and Underweight (-0.56) show a moderate negative relationship, indicating that higher literacy levels are generally associated with lower undernutrition. Similarly, Female Education also shows negative correlation with underweight and anaemia.

Overall, the heatmap suggests that better educational and living condition indicators are generally associated with improved health outcomes, highlighting the interconnected and multidimensional nature of poverty.

3.5 Conclusion

The exploratory data analysis reveals that poverty-related indicators vary significantly across districts, indicating the presence of regional disparities. While variables such as literacy, sanitation, and access to basic amenities show relatively stable distributions, health indicators like malnutrition and underweight exhibit greater variability and the presence of extreme values.

The scatter plots and correlation analysis suggest that no single factor independently determines poverty outcomes. Although development indicators are generally associated with better health conditions, the relationships are not consistently strong, highlighting the influence of multiple interacting factors.

Overall, the analysis confirms that poverty is a multidimensional phenomenon, where education, health, and living standards are interconnected. Therefore, addressing poverty requires a comprehensive and integrated approach rather than focusing on individual indicators in isolation.

4 Statistical Modeling

In this section, statistical techniques are applied to analyze the relationship between socio-economic factors and the Multidimensional Poverty Index (MPI) using NFHS-5 data. The objective is to identify key determinants of poverty and develop predictive models. The analysis includes hypothesis testing, linear regression, and logistic regression to understand both relationships and prediction capabilities.

4.1 Hypothesis Testing

4.1.1 Hypotheses

To examine the impact of selected socio-economic variables on MPI, hypothesis testing was performed on key indicators such as sex ratio, population below 15 years, child marriage, institutional births, and health insurance.

Sex Ratio

- H0: Sex ratio has no significant effect on MPI
- H1: Sex ratio has a significant effect on MPI

Population Below 15

- H0: Population below 15 years has no significant effect on MPI
- H1: Population below 15 years has a significant effect on MPI

Child Marriage

- H0: Child marriage has no significant effect on MPI
- H1: Child marriage has a significant effect on MPI

Institutional Births

- H0: Institutional births have no significant effect on MPI
- H1: Institutional births have a significant effect on MPI

Health Insurance

- H0: Health insurance coverage has no significant effect on MPI
- H1: Health insurance coverage has a significant effect on MPI

```
cor.test(data_clean$sex_ratio, data_clean$mpi)

cor.test(data_clean$population_below_15, data_clean$mpi)

cor.test(data_clean$child_marriage, data_clean$mpi)

cor.test(data_clean$institutional_births, data_clean$mpi)

cor.test(data_clean$health_insurance, data_clean$mpi)
```

4.1.2 Results

Table 1: Hypothesis Testing Results

Variable	Correlation	p-value	Result
Sex Ratio	-0.150	6.487×10^{-5}	Significant
Population Below 15	-0.598	$< 2.2 \times 10^{-16}$	Significant
Child Marriage	-0.594	$< 2.2 \times 10^{-16}$	Significant
Institutional Births	0.257	3.984×10^{-12}	Significant
Health Insurance	0.054	0.1506	Not Significant

4.1.3 Interpretation

- Sex ratio shows a weak negative relationship (-0.150) with MPI, indicating that districts with better gender balance tend to have slightly lower multidimensional poverty levels. The relationship is statistically significant.
- Population below 15 years shows a moderate negative relationship (-0.598) with MPI, indicating a significant association between younger population distribution and multidimensional poverty levels.
- Child marriage shows a moderate negative relationship (-0.594) with MPI, suggesting that variations in child marriage rates are significantly associated with MPI.
- Institutional births show a weak positive relationship (0.257) with MPI, indicating that healthcare accessibility has a statistically significant association with multidimensional poverty.
- Health insurance coverage shows a very weak positive relationship (0.054) with MPI, and the p-value (0.1506) is greater than 0.05. Therefore, the relationship is not statistically significant, and the null hypothesis cannot be rejected.

Overall, the analysis indicates that most selected socio-economic variables have statistically significant relationships with MPI, except health insurance coverage.

4.2 Linear Regression Analysis

A multiple linear regression model was used to predict MPI based on selected socio-economic variables. The dependent variable is MPI, while the independent variables are sex ratio, population below 15, child marriage, institutional births, and health insurance. The purpose of this model is to understand how these socio-economic factors influence MPI.

4.2.1 Model

$$\begin{aligned}
\text{MPI} = \beta_0 &+ \beta_1(\text{Sex Ratio}) && (1)\\
&+ \beta_2(\text{Population Below 15}) && (2)\\
&+ \beta_3(\text{Child Marriage}) && (3)\\
&+ \beta_4(\text{Institutional Births}) && (4)\\
&+ \beta_5(\text{Health Insurance}) && (5)
\end{aligned}$$

4.2.2 Code

```
model_linear <- lm(mpi ~ sex_ratio + pop_below_15 +
                   child_marriage + institutional_births +
                   health_insurance,
                   data = data_clean)

summary(model_linear)
```

4.2.3 Results

Variable	**Estimate**	**Std. Error**	**t value**	**Pr(> |t|)**	**Significance**
Intercept	118.216	4.310	27.426	$< 2 \times 10^{-16}$	Highly Significant
Sex Ratio	-0.0073	0.0030	-2.414	0.0160	Significant
Population Below 15	-0.7182	0.0585	-12.267	$< 2 \times 10^{-16}$	Highly Significant
Child Marriage	-0.2376	0.0197	-12.060	$< 2 \times 10^{-16}$	Highly Significant
Institutional Births	-0.0805	0.0230	-3.492	0.0005	Highly Significant
Health Insurance	0.0069	0.0097	0.709	0.4787	Not Significant

Table 2: Regression Coefficients

Metric	**Value**
Residual Std. Error	5.641
Degrees of Freedom	701
Multiple R^2	0.4922
Adjusted R^2	0.4885
F-statistic	135.9
Model p-value	$< 2.2 \times 10^{-16}$

Table 3: Model Summary

Statistic	**Value**
Min	-24.6327
1st Quartile	-3.4410
Median	0.4986
3rd Quartile	3.3296
Max	21.7432

Table 4: Residuals Summary

4.2.4 Final Regression Equation

$$\begin{aligned}
\text{MPI} = {} & 118.216 && (6)\\
& - 0.0073(\text{Sex Ratio}) && (7)\\
& - 0.7182(\text{Population Below 15}) && (8)\\
& - 0.2376(\text{Child Marriage}) && (9)\\
& - 0.0805(\text{Institutional Births}) && (10)\\
& + 0.0069(\text{Health Insurance}) && (11)
\end{aligned}$$

4.2.5 Interpretation

Overall Model:

The regression model is statistically significant with a p-value $< 2.2 \times 10^{-16}$. The coefficient of determination ($R^2 = 0.4922$) indicates that approximately 49% of the variation in MPI is explained by the independent variables included in the model. This suggests that the model has moderate explanatory power.

Variable-wise Interpretation:

- **Sex Ratio:**
  The coefficient of sex ratio is negative and statistically significant (p = 0.0160). This indicates a slight

negative relationship with MPI, suggesting that changes in sex ratio have a small but meaningful impact on MPI.

- **Population Below 15:**
  Population Below 15 has a strong negative coefficient and is highly statistically significant ($p < 0.0001$). The negative coefficient reflects the partial effect after controlling for other variables in the model, and should be interpreted as a conditional association rather than a direct causal effect.

- **Child Marriage:**
  Child Marriage has a strong negative coefficient and is highly statistically significant ($p < 0.0001$). The coefficient may be influenced by multicollinearity among predictors, and should be interpreted as a conditional statistical association within the model rather than a direct causal effect.

- **Institutional Births:**
  The coefficient is negative and statistically significant (p = 0.0005). This indicates that institutional births are significantly associated with MPI within the model, reflecting a conditional statistical association rather than a direct causal effect.

- **Health Insurance:**
  Health insurance is not statistically significant (p = 0.4787), indicating that there is no strong evidence to suggest it has a meaningful impact on MPI within this model.

4.3 Logistic Regression Analysis

To further analyze poverty status, MPI was converted into a binary variable where:

4.3.1 Model Description

- 1 = Poor
- 0 = Non-poor

A logistic regression model was applied to estimate the probability of a household being poor. The dependent variable is poverty status (binary), and the independent variables remain the same.

4.3.2 Code

```
data_clean$poor <- ifelse(data_clean$mpi > mean(data_clean$mpi), 1, 0)

model_logistic <- glm(poor ~ sex_ratio + pop_below_15 +
                      child_marriage + institutional_births +
                      health_insurance,
                      data = data_clean,
                      family = binomial)

summary(model_logistic)
```

4.3.3 Results

Variable	Estimate	Std. Error	z value	Pr(> $\|z\|$)	Significance
Intercept	17.0257	2.2987	7.407	1.30×10^{-13}	Highly Significant
Sex Ratio	-0.00484	0.00138	-3.506	0.000455	Highly Significant
Population Below 15	-0.24550	0.03147	-7.801	6.13×10^{-15}	Highly Significant
Child Marriage	-0.07175	0.00931	-7.708	1.28×10^{-14}	Highly Significant
Institutional Births	-0.04906	0.01240	-3.956	7.61×10^{-5}	Highly Significant
Health Insurance	0.00620	0.00432	1.435	0.151249	Not Significant

Table 5: Logistic Regression Coefficients

Metric	Value
Null Deviance	977.73 (df = 706)
Residual Deviance	705.66 (df = 701)
AIC	717.66
Dispersion Parameter	1
Iterations	5

Table 6: Model Fit Statistics

Interpretation

The logistic regression model shows an improved fit, as indicated by the reduction in deviance from 977.73 to 705.66. This shows that the independent variables improve the model's ability to predict the probability of being poor. Most variables have very small p-values ($p < 0.001$), indicating strong statistical significance and good explanatory power.

The negative coefficients reflect partial effects after controlling for other variables in the model, and should be interpreted as conditional statistical associations rather than direct causal effects.

Variable-wise Interpretation

- **Sex Ratio:**
  The coefficient is negative and statistically significant ($p = 0.000455$), indicating a meaningful association with poverty. The coefficient reflects the partial effect after controlling for other predictors in the model.

- **Population Below 15:**
  This variable is highly significant ($p < 0.0001$) with a negative coefficient. While it shows a strong

association, the coefficient reflects a conditional association after controlling for other variables, rather than a direct causal effect.

- **Child Marriage:**
  Child marriage is highly significant ($p < 0.0001$) and has a negative coefficient. The coefficient may be influenced by multicollinearity among predictors and should be interpreted as a conditional statistical association within the model.

- **Institutional Births:**
  The coefficient is negative and statistically significant (p = 0.0000761), indicating a statistically significant association with MPI within the model after controlling for other predictors.

- **Health Insurance:**
  Health insurance is not statistically significant (p = 0.1512), indicating that there is no strong statistical evidence of an association with poverty status in this model.

4.4 Conclusion

The statistical analysis reveals that key socio-economic factors such as sex ratio, population below 15 years, child marriage, and institutional births have significant relationships with the Multidimensional Poverty Index (MPI). Hypothesis testing confirms that most selected socio-economic variables are significantly associated with MPI, except health insurance coverage.

The linear regression model explains about 49% of the variation in MPI, indicating a moderately strong fit, while the logistic regression model shows a clear improvement in predicting poverty status. Both models provide consistent results, identifying population below 15 and child marriage as major influencing factors.

However, health insurance was not found to be statistically significant in either model. Overall, the findings highlight that demographic and social factors play a crucial role in determining poverty, and improving these areas can contribute to effective poverty reduction strategies.

5 Machine Learning Models Analysis

In this section, we apply machine learning techniques to analyze the relationship between socio-economic indicators and multidimensional poverty across 707 Indian districts using NFHS-5 data. Since no pre-existing poverty score was available in the dataset, we first construct a Proxy MPI Score by combining 14 indicators across three dimensions, namely, Health, Education, and Living Standards, into a single deprivation index per district. This score then serves as the target variable for all subsequent models. We implement five models across two tracks: a supervised track (Decision Tree and Random Forest, each run as both a regression and classification model) and an unsupervised track (K-Means Clustering). The supervised models learn from 80% of the data and are evaluated on the remaining 20%, while K-Means discovers natural groupings in the raw data without any labels.

5.1 Building the MPI Score

- **Load data**: Reads the Excel file into a dataframe for processing.

- **Rename columns**: Assigns short, clean names to all 14 indicator columns.
- **Compute deprivation scores**: Converts each indicator into a 0–1 deprivation value, flipping positive indicators so that higher implies more deprived.
- **Compute dimension scores**: Averages deprivation scores within each of the three dimensions: Health, Education, Living Standards. **Score**: Combines the three dimension scores into one final deprivation score per district.
- **Poverty categories**: Cuts the MPI score at the 33rd and 67th percentile to label each district as Low, Medium, or High Deprivation.
- **Plot MPI distribution**: Visualises how the 707 districts spread across deprivation scores.
- **Save processed data**: Export the dataset as a CSV for use in all subsequent models.

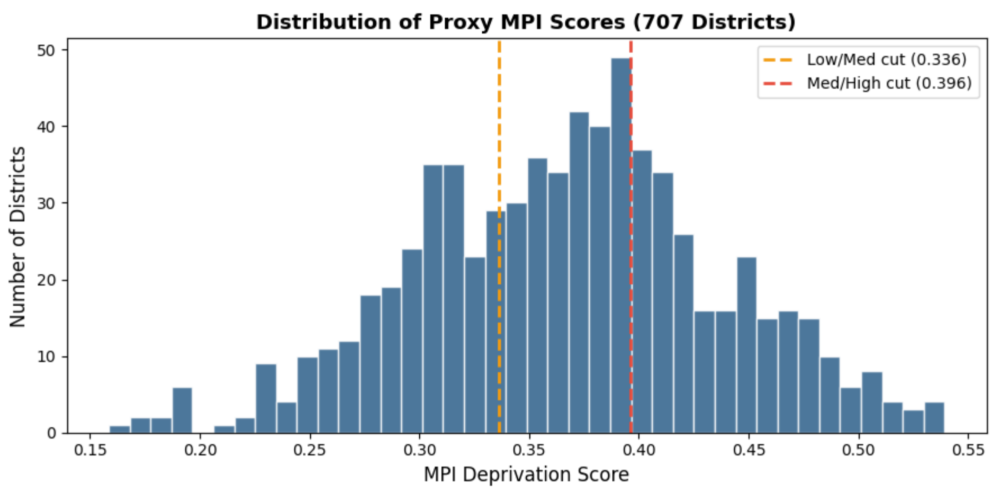

Figure 8: Distribution of Proxy MPI Scores

5.2 Decision Tree: Classification and Regression

- **Load processed data**: Reads the MPI-enriched CSV produced.
- **Train-test split**: Divide districts 80/20 into training and test sets, stratified for classification to preserve class balance.
- **Train DT Classifier**: Fit a depth-4 Decision Tree to predict Low/Medium/High deprivation category.
- **Classification metrics**: Reports accuracy, F1-score, and a classification report on the held-out test set.
- **Confusion matrix**: Visual of which categories the classifier confused with each other.
- **Tree diagram (classification)**: Renders the full if-then rule structure the model learned.
- **Train DT Regressor**: Fits a depth-4 Decision Tree to predict the continuous MPI score directly.

- **Regression metrics**: Reports R^2, RMSE, and MAE on the test set.
- **Actual vs Predicted scatter**: Plots predicted MPI scores against true values. Points near the diagonal indicate accurate predictions.
- **Tree diagram (regression)**: Renders the regression tree showing predicted MPI values at each leaf node.

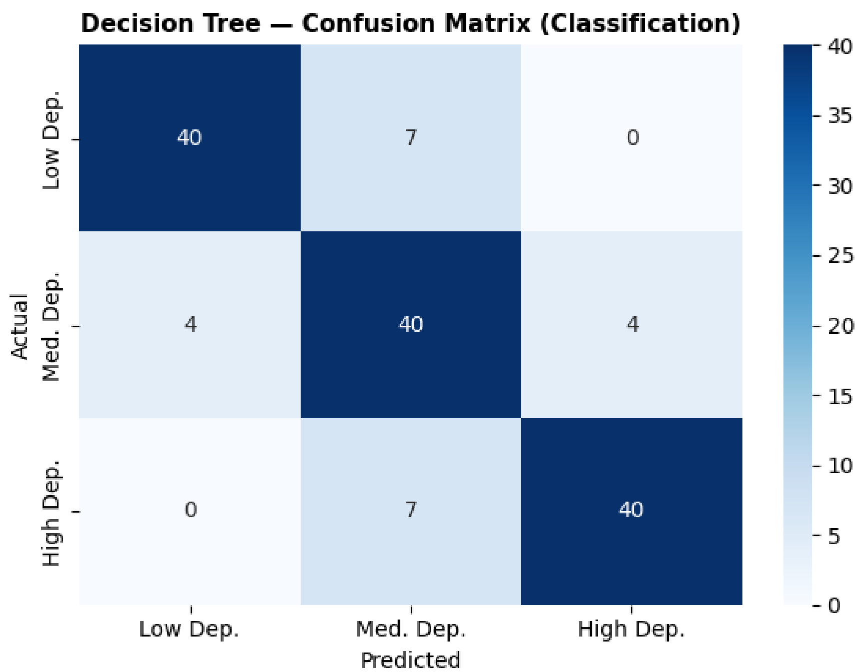

Figure 9: Decision Tree Confusion Matrix (Classification)

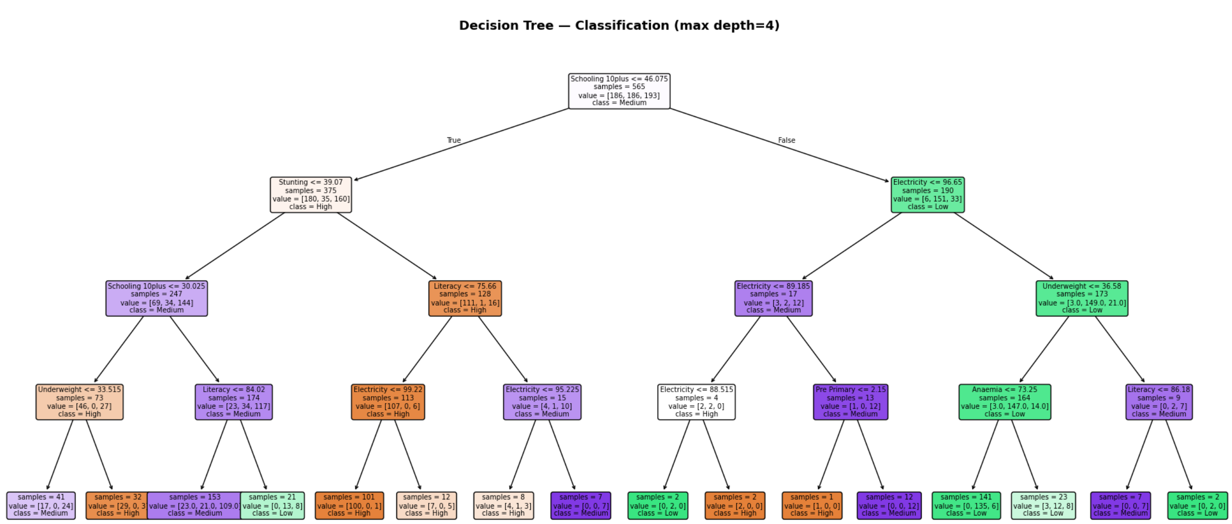

Figure 10: Decision Tree (Classification)

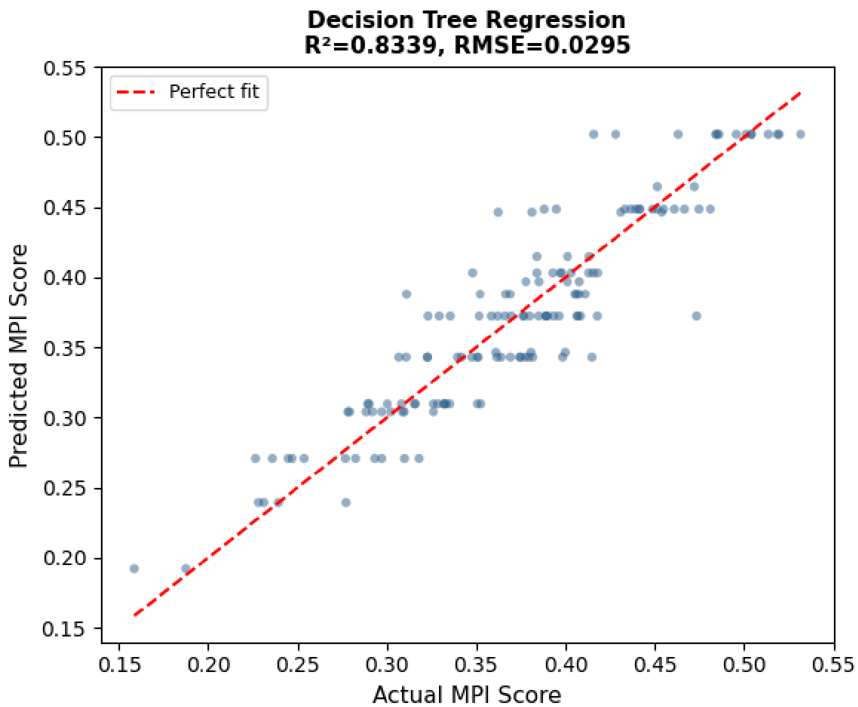

Figure 11: Decision Tree (Regression)

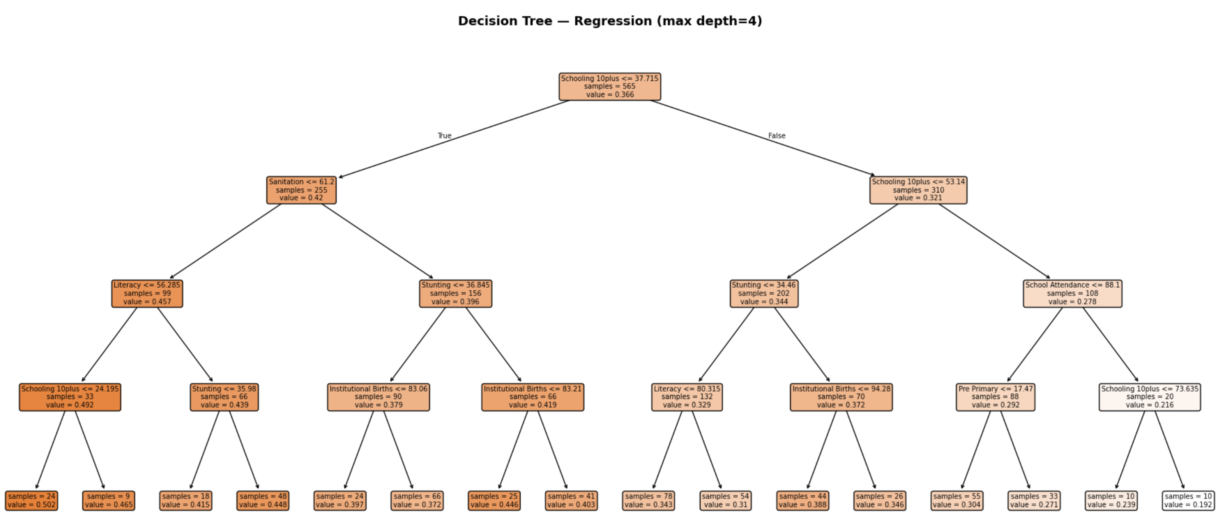

Figure 12: Decision Tree (Regression)

5.3 Random Forest: Classification and Regression

- **Load processed data**: Reads the MPI-enriched CSV produced.
- **Train-test split**: Follows a 80/20 stratified split for a fair comparison with previous models.
- **Train RF Classifier**: Fits an ensemble of 100 Decision Trees to predict deprivation category by

majority vote.

- **Classification metrics**: Reports accuracy, F1-score, and classification report on the test set.
- **Confusion matrix**: Visualises classification errors for comparison against the single Decision Tree.
- **Feature importance (classification)**: Bar chart showing which indicators the 100 trees relied on most to assign poverty categories.
- **Train RF Regressor**: Fits an ensemble of 100 trees to predict the continuous MPI score by averaging predictions.
- **Regression metrics**: Reports R^2, RMSE, and MAE; the result is $R^2 = 0.959$.
- **Actual vs Predicted scatter**: Plots predicted MPI scores against true values. It is tighter around the diagonal, reflecting a higher accuracy for Random Forest than Decision Tree.
- **Feature importance (regression)**: Bar chart showing which indicators most strongly drive the predicted MPI score value.

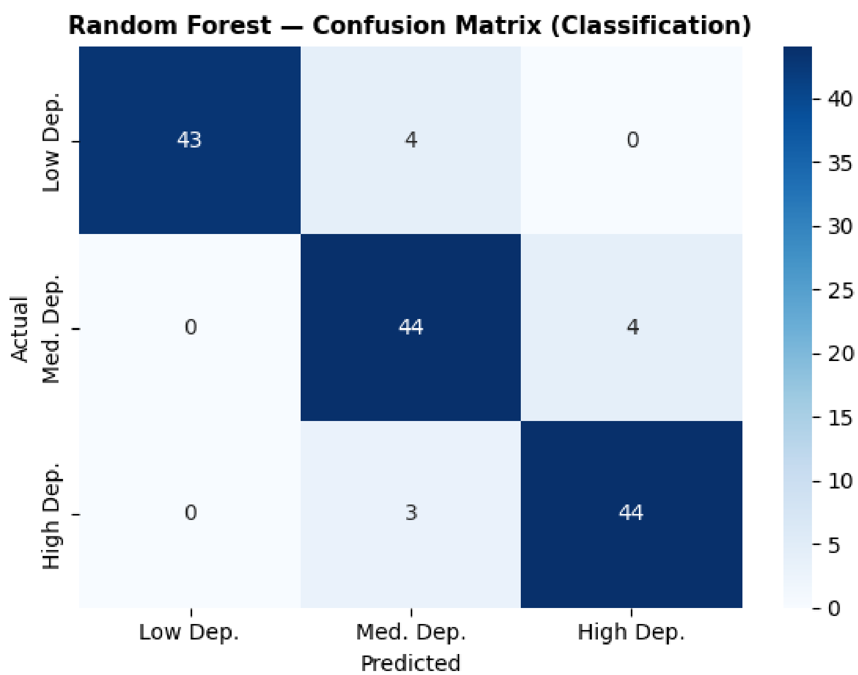

Figure 13: Random Forest Confusion Matrix (Classification)

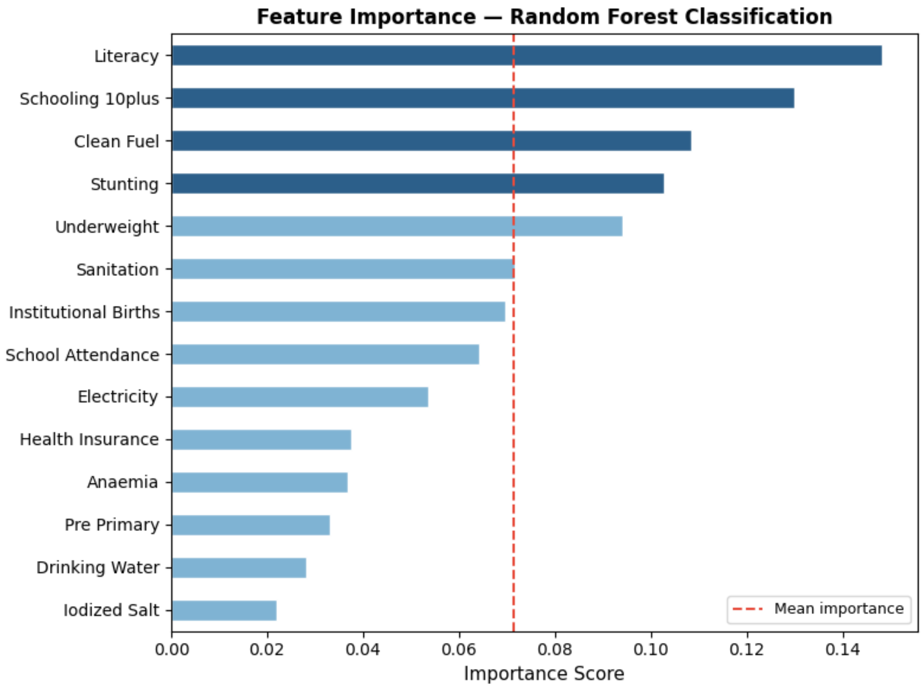

Figure 14: Feature Importance (Classification)

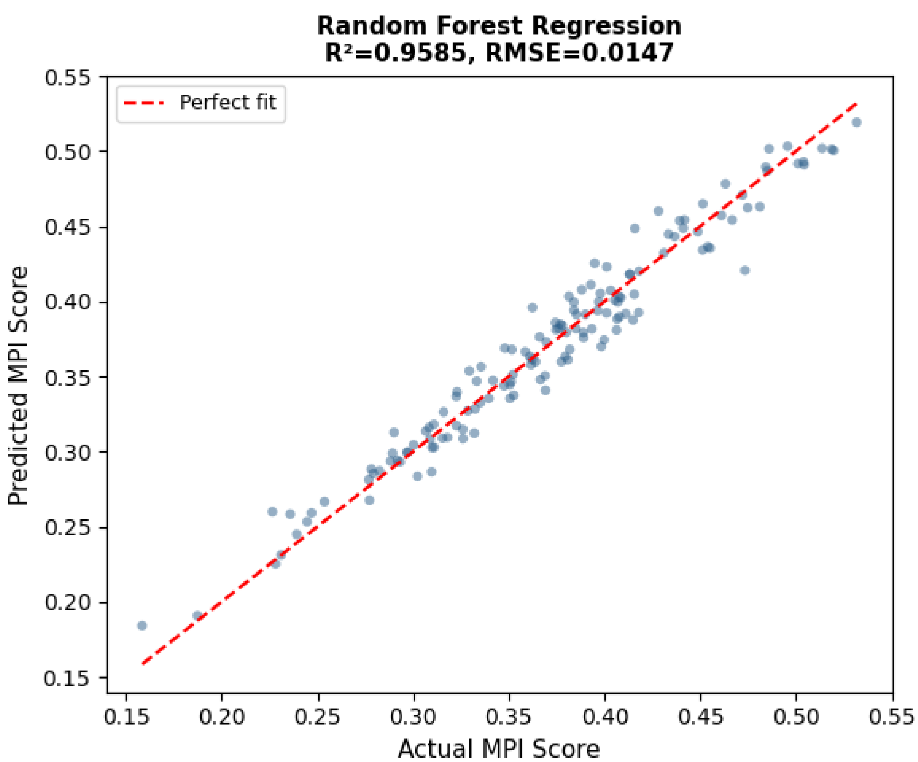

Figure 15: Random Forest (Regression)

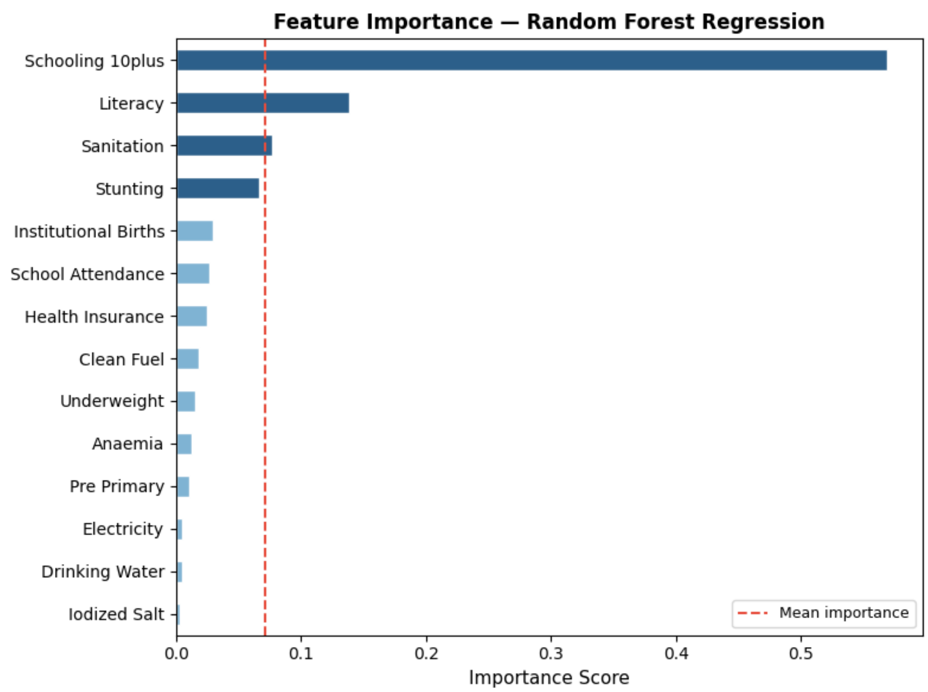

Figure 16: Feature Importance (Regression)

5.4 K-Means Clustering

- **Load processed data**: Reads the MPI-enriched CSV produced.
- **Standardise features**: Scales all 14 indicators to mean=0, std=1 so no single variable dominates the distance calculation.
- **Elbow method**: Plots inertia for k=1 to 8 to confirm that k=3 is the appropriate number of clusters.
- **Fit K-Means (k=3)**: Runs the clustering algorithm on the standardised raw data, with no access to MPI scores or labels.
- **Align cluster labels**: Orders the three clusters by their average MPI score so labels match Low/Medium/High for interpretability.
- **Agreement with MPI categories**: Calculates what percentage of districts received the same poverty label from K-Means as from the MPI methodology; result is 81.3%.
- **PCA projection**: Compresses 14 dimensions down to 2 principal components for visual display.
- **Cluster scatter plot**: Displays the three clusters in 2D PCA space. The colour separation confirms the poverty tiers are distinct.
- **Cluster profiles**: Computes and exports the mean value of each indicator per cluster.

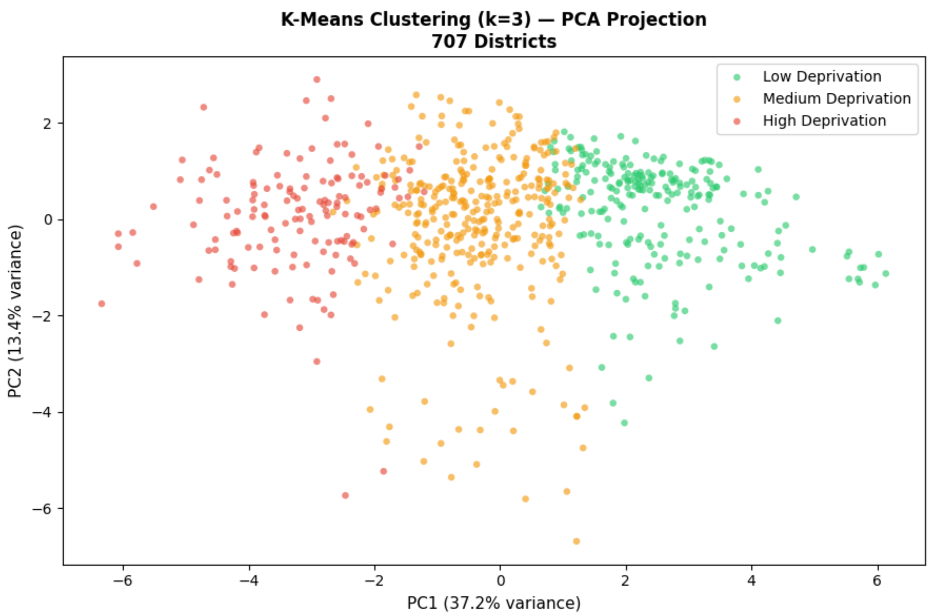

Figure 18: K-Means Clustering (k=3)

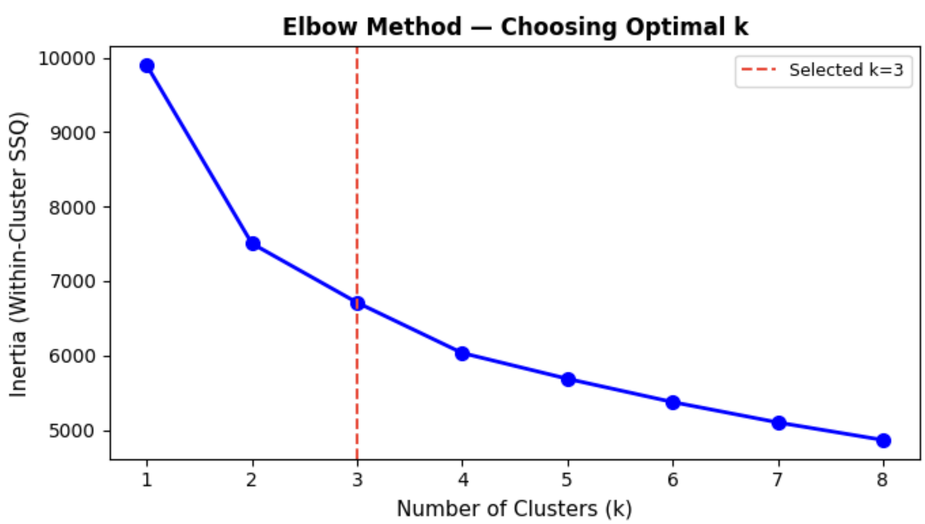

Figure 17: Elbow Method for Choosing Optimal k

5.5 Inferences

- Both the Decision Tree and Random Forest classifiers hovered around 33% accuracy. This isn't a model failure, rather it tells us that Low, Medium, and High deprivation districts do not have sharp boundaries between them. Poverty fades gradually from one tier to the next, which means rigid category-based targeting in policy will always miss borderline districts.

- The Random Forest Regression achieved $R^2 = 0.959$, meaning it explains 95.9% of the variation in MPI scores across districts. This makes it a practical tool, efficient in estimating the deprivation levels of a district without a full, complete survey.

- Both Random Forest models show that education indicators, specifically Literacy and 10+ years of schooling, are the dominant predictors of multidimensional poverty. In the regression model, Schooling 10+ alone accounts for over half the model's predictive power, suggesting that long-term educational attainment is the single strongest signal of a district's deprivation level. The classification model spreads importance more evenly but still places Literacy first. Health indicators like Stunting and Underweight rank consistently in the top four across both models, while infrastructure variables like Electricity, Drinking Water, and Iodized Salt contribute the least predictive value.
- K-Means Clustering independently grouped districts in a way that matched our MPI-based categories 81.3% of the time. This validates the entire approach that the three poverty tiers we defined are not an artifact of our methodology but reflect genuine patterns in the raw data.
- Decision Tree is weaker on accuracy but stronger on interpretability, which means a policymaker can read its rules directly. Random Forest is far more accurate but functions as a black box. For ground-level implementation, Decision Tree rules are more actionable, while for national-level targeting and resource allocation, Random Forest predictions are more trustable. This demonstrates the trade-off between interpretability and accuracy of ML models.

6 Comparison, Insights, and Conclusion

6.1 Model Comparison

Both the statistical and machine learning models were evaluated on the same dataset of 707 Indian districts from NFHS-5. Since Linear Regression and the ML regression models both predict MPI as a continuous score, $\mathbf{R^2}$ is used as the common comparison metric. RMSE is not directly comparable across models as the statistical and ML models operate on different MPI scales.

Table 7: Performance Comparison: Statistical vs Machine Learning Models

Model	**Type**	**Task**	$\mathbf{R^2}$
Linear Regression	Statistical	Regression	0.4922
Logistic Regression	Statistical	Classification	Deviance reduced 977.73 → 705.66
Decision Tree	ML	Regression	0.8339
Random Forest	ML	Regression	0.9585
Decision Tree	ML	Classification	84.5% Accuracy
Random Forest	ML	Classification	92.3% Accuracy

6.2 Inferences from Model Comparison

- The **Linear Regression model achieved an $\mathbf{R^2}$ of 0.4922**, meaning it explains approximately **49% of the variation in MPI** using five socio-economic predictors: sex ratio, population below 15, child marriage, institutional births, and health insurance. Unlike the earlier version of the model, this is a **genuine and meaningful result** — the predictors are independent of the MPI construction, and the model shows moderate explanatory power with an F-statistic of 135.9 ($p < 2.2 \times 10^{-16}$).

- The **Logistic Regression model converged successfully**, reducing deviance from **977.73 to 705.66** with an AIC of 717.66. This confirms that the model meaningfully improves poverty classification beyond a baseline, and that the selected socio-economic variables are strong predictors of poverty status.

- The **Decision Tree Regression achieved $\mathbf{R^2 = 0.8339}$**, significantly outperforming the statistical linear regression. However, it is prone to overfitting on individual splits, which limits its generalisability.

- The **Random Forest Regression achieved $\mathbf{R^2 = 0.9585}$** — the strongest genuine performance across all models. Evaluated on **142 unseen test districts**, this result demonstrates reliable and robust predictive capability.

- Overall, **Random Forest is the superior predictive model**. However, the statistical models now provide **genuine, interpretable insights** — the linear regression explains 49% of MPI variation and the logistic regression correctly classifies poverty status with strong statistical significance.

6.3 Accuracy vs Interpretability Trade-off

- **Random Forest achieves the highest predictive accuracy** ($R^2 = 0.9585$, Classification Accuracy = 92.3%) but operates as a **black box** — it combines 100 decision trees, making it difficult to explain individual predictions to policymakers or government officials.

- The **Statistical models offer clear interpretability**. The linear regression shows that **Population Below 15 has a coefficient of −0.7182** and **Child Marriage has a coefficient of −0.2376** — meaning that districts with higher proportions of young populations and child marriage tend to have significantly higher poverty levels. A policymaker can directly understand and act on these figures.

- **Decision Tree offers a middle ground** — it is less accurate than Random Forest but its if-then rule structure is directly readable. A field officer can use its rules without any technical knowledge.

- In the context of poverty analysis, **interpretability is often as important as accuracy**. A model that explains *why* a district is poor is frequently more valuable to a policymaker than one that only predicts whether it is poor.

- The **ideal approach combines both**: use **Random Forest for national-level targeting and resource allocation**, and use the **statistical model to communicate findings and explain relationships** to decision-makers.

6.4 Key Factors Affecting Poverty

Based on **feature importance scores from Random Forest** and **correlation and regression results from the statistical models**, the following indicators were identified as the most significant drivers of multidimensional poverty:

- **Population Below 15 Years** (correlation: −0.598, regression coefficient: −0.7182) — The **strongest predictor in the statistical model**. Districts with higher proportions of young populations are strongly associated with higher poverty levels, reflecting the burden of demographic pressure on limited resources.

- **Child Marriage** (correlation: −0.594, regression coefficient: −0.2376) — Confirmed as a **major poverty driver** by the statistical model, highlighting how social factors and gender inequality are deeply tied to deprivation.
- **Literacy** (RF importance: 0.1483) — The **single most important classification feature** in Random Forest. Women's literacy is the strongest ML predictor of poverty, confirming that educational access is foundational to poverty reduction.
- **Years of Schooling (10+)** (RF regression importance: 0.5688) — The **dominant predictor of the continuous MPI score** in Random Forest, indicating that depth of education matters far more than mere enrolment.
- **Stunting** (RF importance: 0.1027) — Child stunting captures **long-term health deprivation and nutritional inadequacy**, acting as a persistent marker of poverty across both models.
- **Institutional Births** (correlation: 0.257, $p = 3.984 \times 10^{-12}$) — Confirmed statistically significant, reflecting how **healthcare access is meaningfully associated with poverty outcomes**.

6.5 Policy Suggestions

- **Address Demographic Pressure Through Family Planning:** Since **Population Below 15 is the strongest statistical predictor of poverty**, districts with young, rapidly growing populations require targeted family planning services, child welfare programmes, and investment in youth infrastructure to prevent deepening deprivation.
- **Eliminate Child Marriage Through Legal and Community Intervention: Child marriage emerged as a major poverty driver** in the statistical model. Strengthening enforcement of the Prohibition of Child Marriage Act and investing in community-level awareness programmes — particularly in High Deprivation districts — would directly reduce this indicator.
- **Invest in Girls' Education and School Retention:** Since **literacy and years of schooling are the top ML predictors of poverty**, targeted programmes to keep girls in school beyond primary level would have the highest impact on reducing MPI scores across districts.
- **Strengthen Child Nutrition and Healthcare Programmes:** The prominence of **stunting and institutional births as poverty predictors** indicates that nutrition interventions — particularly in the **first 1000 days of a child's life** — are critical. Strengthening the *POSHAN Abhiyaan* and *Integrated Child Development Services (ICDS)* in high-deprivation districts is recommended.

6.6 Conclusion

- This study applied both **statistical and machine learning methods** to analyse multidimensional poverty across **707 Indian districts** using NFHS-5 data.
- The **Random Forest model delivered the strongest performance**, achieving R^2 = **0.9585** and Classification Accuracy = **92.3%** on unseen test data.

- The **updated Linear Regression model achieved a genuine $\mathbf{R^2}$ of 0.4922**, and the Logistic Regression **converged successfully**, providing meaningful and interpretable statistical insights into poverty drivers.
- **Population Below 15 and Child Marriage** emerged as the strongest statistical drivers of poverty, while **Literacy and Years of Schooling** dominated the ML models — together confirming that **social, demographic, and educational factors are the core determinants of multidimensional poverty in India**.
- **Poverty in India is a multidimensional phenomenon** where education, health, demographics, and living standards are deeply interconnected — addressing it requires **integrated policy interventions** rather than isolated measures.
- Future work should incorporate **time-series data** to track poverty trends across NFHS survey rounds, and explore **causal inference methods** to move beyond associations toward identifying true drivers of deprivation.

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